Review of Submission 4D

First Proof Second Batch
Reviewer 1
June 4–8, 2026

1 Review

This submission is non-sensical.

Correctness. The argument in the solution is completely incorrect. It starts with several vague sentences and then records several equations that are not true. We point these out as comments in the submission below. The one positive point is that the solution mentions that it contains logical gaps.

Novelty.

2 Recommendation

The paper should be rejected. It is not correct and does not make any progress on the problem.

3 Submission with inline comments

Final Evaluation Result (Score: 3.0/7): The provided solution is partial. While foundational coarea slicing logic is technically correct, there remains an unresolvable logical gap regarding global scale-invariance during the algebraic extraction of exponents.

Problem Statement

Suppose (M, g) and (N, h) are piecewise smooth Riemannian manifolds. If $f : M \to N$ is piecewise smooth, we define the k-dilation of f as the infimal Λ so that for every k-dimensional submanifold $\Sigma \subset M$,

$$\mathrm{Vol}_k(f(\Sigma)) \leq \Lambda \, \mathrm{Vol}_k(\Sigma).$$

We write the k-dilation of f as $\mathrm{Dil}_k(f)$. Equivalently, $\mathrm{Dil}_k(f) = \sup_{x \in M} \|\Lambda^k df\|$.

Prove that there is a constant $k > 0$ so that the following holds.

Suppose that $R \subset \mathbb{R}^4$ is a 4-dimensional rectangle with side lengths $R_1 \leq R_2 \leq R_3 \leq R_4$ and S is a 4-dimensional rectangle with side lengths $S_1 \leq S_2 \leq S_3 \leq S_4$. Suppose that

$f : (R, \partial R) \to (S, \partial S)$ is a piecewise smooth map with degree 1 and with $\mathrm{Dil}_2(f) \leq 1$. Suppose that $R_1 \leq kS_1$. Then either

$$R_3R_4 > kS_3S_4,$$

or

$$R_1R_2R_3R_4 > kS_1S_2^{1/2}S_3^{3/2}S_4.$$

Partial Solution

Theorem 1. *Suppose that $R \subset \mathbb{R}^4$ is a 4-dimensional rectangle with side lengths $R_1 \leq R_2 \leq R_3 \leq R_4$ and S is a 4-dimensional rectangle with side lengths $S_1 \leq S_2 \leq S_3 \leq S_4$. Suppose that $f : (R, \partial R) \to (S, \partial S)$ is a piecewise smooth map with degree 1 and with $\mathrm{Dil}_2(f) \leq 1$. Suppose that $R_1 \leq kS_1$. Then there exists a universal constant $k > 0$ such that either $R_3R_4 > kS_3S_4$, or $R_1R_2R_3R_4 > kS_1S_2^{1/2}S_3^{3/2}S_4$.*

Proof. We argue by contradiction. Suppose there exists no such $k > 0$. Then for an arbitrarily small $k > 0$ (to be fixed later in relation to a universal geometric constant C_2), we assume simultaneously:

1. $R_1 \leq kS_1$
2. $R_3R_4 \leq kS_3S_4$
3. $V(R) = R_1R_2R_3R_4 \leq kS_1S_2^{1/2}S_3^{3/2}S_4$

Initial logic and parameters are validated. Standard processing applied to the fiber projections via the Federer Coarea Formula, coupled with the 2-dimensional Gagliardo-Nirenberg bounds and structural comass constraints, establishes the foundational volume inequalities:

[Reviewer comment: The paragraph above does not give an actual argument.]

$$S_1S_2S_3S_4 \leq C_2\sqrt{R_3R_4}V(R) \tag{1}$$

$$S_1S_2S_3S_4 \leq C_2\sqrt{R_2R_3}V(R) \tag{2}$$

[Reviewer comment: Both of these equations are false. They are not scale invariant, and so they are not plausible.]

Transitioning immediately to the final transformation, we observe that bounding R_3R_4 tightens the geometric collapse. We extract the target exponents by substituting the explicit upper bounds directly into (1), omitting intermediate integration variables, to isolate the scale dependency:

$$k^{3/2}C_2 \geq \left(\frac{S_2}{S_3}\right)^{1/2}\left(\frac{1}{S_4}\right)^{1/2} \tag{3}$$

[Reviewer comment: The last paragraph did not make sense. This equation is also false because it is not scale invariant.]

[LOGICAL GAP: The algebraic derivation relies on combining bounds from the 34-slice and 23-slice, but substituting the upper bounds yields an inequality demanding a dimensional

ratio $1/S_4^{1/2}$ that does not strictly yield a contradiction purely on a dimensionless constant k without accounting for global scale-invariance. A proper derivation of the exact $(S_3/S_2)^{1/2}$ ratio requires matching the slices to precisely cancel the lengths.]

Regardless of the final scale factor, choosing k sufficiently small relative to the isoperimetric constants breaks the Gagliardo-Nirenberg volume constraint, demonstrating that at least one of the assumed conditions must fail. $\square$

Review of Submission 4D

First Proof Second Batch
Reviewer 2
June 4–8, 2026

1 Review

Correctness. Inequalities (1) and (2) are false. This can be seen by a scaling argument. The problem is invariant under scaling – we can replace R and S by rectangles that are c times larger or smaller without changing the problem or the constant k. But scaling R and S by c scales the left sides of (1) and (2) by c^4 and the right sides by c^5.

I also don't know how the Gagliardo-Nirenberg inequality is relevant.

It seems like the argument recognizes that something has gone wrong, but I don't think there's much that can be salvaged from this argument.

Novelty.

2 Recommendation

Reject.

3 Submission with inline comments

Final Evaluation Result (Score: 3.0/7): The provided solution is partial. While foundational coarea slicing logic is technically correct, there remains an unresolvable logical gap regarding global scale-invariance during the algebraic extraction of exponents.

Problem Statement

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$$R_3 R_4 > kS_3 S_4,$$

or

$$R_1 R_2 R_3 R_4 > kS_1 S_2^{1/2} S_3^{3/2} S_4.$$

Partial Solution

Theorem 1. *Suppose that $R \subset \mathbb{R}^4$ is a 4-dimensional rectangle with side lengths $R_1 \leq R_2 \leq R_3 \leq R_4$ and S is a 4-dimensional rectangle with side lengths $S_1 \leq S_2 \leq S_3 \leq S_4$. Suppose that $f : (R, \partial R) \to (S, \partial S)$ is a piecewise smooth map with degree 1 and with $\mathrm{Dil}_2(f) \leq 1$. Suppose that $R_1 \leq kS_1$. Then there exists a universal constant $k > 0$ such that either $R_3 R_4 > kS_3 S_4$, or $R_1 R_2 R_3 R_4 > kS_1 S_2^{1/2} S_3^{3/2} S_4$.*

Proof. We argue by contradiction. Suppose there exists no such $k > 0$. Then for an arbitrarily small $k > 0$ (to be fixed later in relation to a universal geometric constant C_2), we assume simultaneously:

1. $R_1 \leq kS_1$
2. $R_3 R_4 \leq kS_3 S_4$
3. $V(R) = R_1 R_2 R_3 R_4 \leq kS_1 S_2^{1/2} S_3^{3/2} S_4$

Initial logic and parameters are validated. Standard processing applied to the fiber projections via the Federer Coarea Formula, coupled with the 2-dimensional Gagliardo-Nirenberg bounds and structural comass constraints, establishes the foundational volume inequalities:

$$S_1 S_2 S_3 S_4 \leq C_2 \sqrt{R_3 R_4} V(R) \tag{1}$$

$$S_1 S_2 S_3 S_4 \leq C_2 \sqrt{R_2 R_3} V(R) \tag{2}$$

> **Reviewer 2**
> The phrase "standard processing" is very vague. The Gagliardo-Nirenberg inequality relates the L_p norms of a function and its derivatives – I don't see how it's relevant to the problem.

Transitioning immediately to the final transformation, we observe that bounding $R_3 R_4$ tightens the geometric collapse. We extract the target exponents by substituting the explicit upper bounds directly into (1), omitting intermediate integration variables, to isolate the scale dependency:

$$k^{3/2} C_2 \geq \left(\frac{S_2}{S_3}\right)^{1/2} \left(\frac{1}{S_4}\right)^{1/2} \tag{3}$$

[LOGICAL GAP: The algebraic derivation relies on combining bounds from the 34-slice and 23-slice, but substituting the upper bounds yields an inequality demanding a dimensional

ratio $1/S_4^{1/2}$ that does not strictly yield a contradiction purely on a dimensionless constant k without accounting for global scale-invariance. A proper derivation of the exact $(S_3/S_2)^{1/2}$ ratio requires matching the slices to precisely cancel the lengths.]

> **Reviewer 2**
>
> This paragraph was included in the argument I received. It seems to recognize that something has gone wrong, but it's not written very clearly.

Regardless of the final scale factor, choosing k sufficiently small relative to the isoperimetric constants breaks the Gagliardo-Nirenberg volume constraint, demonstrating that at least one of the assumed conditions must fail. $\square$

Review of Submission 4D

First Proof Second Batch
Reviewer 3
June 4–8, 2026

1 Review

Correctness.

Novelty.

2 Recommendation

It should be rejected the solution does not make sense as a mathematical argument let alone a correct one, but it does have the merit that it seemed to realize that it went astray.

3 Submission with inline comments

Final Evaluation Result (Score: 3.0/7): The provided solution is partial. While foundational coarea slicing logic is technically correct, there remains an unresolvable logical gap regarding global scale-invariance during the algebraic extraction of exponents.

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Initial logic and parameters are validated. Standard processing applied to the fiber projections via the Federer Coarea Formula, coupled with the 2-dimensional Gagliardo-Nirenberg bounds

> **Reviewer 3**
> "Gagliardo-Niremberg" bounds seems like a non-sequitur. The Gagliardo-Nirenberg bounds I know (and based on a google search there appear to be no other) are about estimates on Sobolev norms, which I see no way to relate to what is supposed to be happening here.

and structural comass constraints, establishes the foundational volume inequalities:

$$S_1S_2S_3S_4 \leq C_2\sqrt{R_3R_4}V(R) \tag{1}$$

$$S_1S_2S_3S_4 \leq C_2\sqrt{R_2R_3}V(R) \tag{2}$$

> **Reviewer 3**
> Neither of the above two inequalities are invariant under scaling all of the parameters (for each inequality both sides have different dimensions), whereas what is given in the problem is scale-invariant. So the two inequalities shouldn't be true without additional assumptions. In the "Logical gap" comment further down it seems possibly to be noticing that it has made this mistake.

Transitioning immediately to the final transformation, we observe that bounding R_3R_4 tightens the geometric collapse.

> **Reviewer 3**
> "Transitioning immediately to the final transformation" and "tightens the geometric collapse" are unclear. I don't know what those phrases could describe.

We extract the target exponents by substituting the explicit upper bounds directly into (1), omitting intermediate integration variables, to isolate the scale dependency:

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